CA-1 ASSIGNMENT

APPLIED STATISTICS & MACHINE LEARNING

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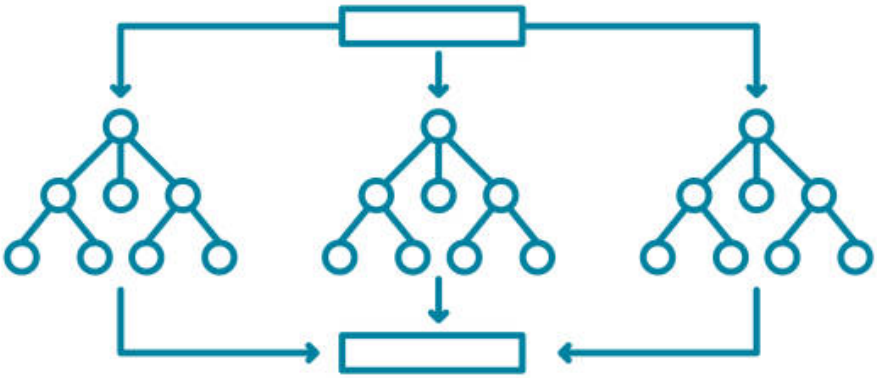

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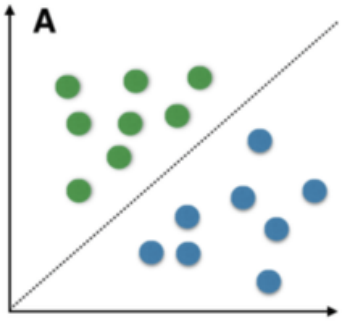

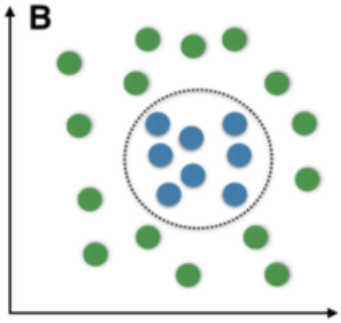

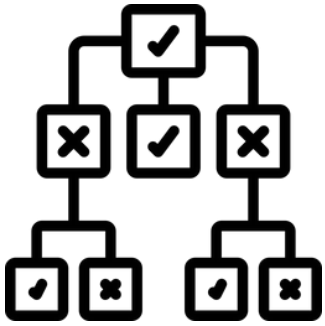

CRITICAL ANALYSIS REPORT

Introduction

The objective of this task is to create a classification model in Python using random forest and support vector classification algorithms to predict customer churn for a bank. The focus is on providing a critical analysis of the approach and results obtained. Customer churn prediction plays a vital role for banks as it enables proactive measures to retain potential churners. The PDF report will detail the methodology employed, including data preprocessing, model implementation, and result evaluation. Additionally, it will discuss the strengths and limitations of the algorithms used, as well as suggest potential improvements for future work.

LET'S UNDERSTAND DATA & VARIABLES

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About Dataset

The dataset CustomerChurn.csv consists of information related to a bank's credit card customers. Each row in the dataset represents a unique customer and contains various details about their profile. These details include attributes such as customer age, marital status, income category, and several others.

Variables Declaration

```
Data columns (total 16 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Attrition_Flag            6237 non-null   object
 1   Customer_Age              6237 non-null   int64
 2   Gender                    6237 non-null   object
 3   Dependent_count           6237 non-null   int64
 4   Education_Level           6237 non-null   object
 5   Marital_Status            6237 non-null   object
 6   Income_Category           6237 non-null   object
 7   Card_Category             6237 non-null   object
 8   Months_on_book            6237 non-null   int64
 9   Total_Relationship_Count  6237 non-null   int64
 10  Months_Inactive           6237 non-null   int64
 11  Contacts_Count            6237 non-null   int64
 12  Credit_Limit              6237 non-null   float64
 13  Total_Revolving_Bal       6237 non-null   int64
 14  Total_Trans_Amt           6237 non-null   int64
 15  Total_Trans_Ct            6237 non-null   int64
dtypes: float64(1), int64(9), object(6)
memory usage: 779.8+ KB
```

Target Variables

```
#   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
0   Attrition_Flag            6237 non-null   object
```

DATASET PREPARATION

04

Data Types & Memory Usage

```
dtypes: float64(1), int64(9), object(6)
memory usage: 779.8+ KB
```

The Data Preprocessing Steps Performed On The Dataset Are As Follows:

- **Checking For Null Values:** The info() function provides a summary of the dataset, including the number of null values for each column. Any columns with null values should be handled before proceeding with the analysis.

- **Converting Categorical Features**: To meet the prerequisites of a classification model, we are undertaking the task of converting the "Gender" and "Attrition_Flag" columns from categorical features into numerical data. This conversion is necessary because classification models are unable to process string values directly.

- **Handling Features With Multiple Categories:** columns, such as Marital_Status,Education_Level,Income_Category,and Card_Category, have multiple categories. These columns can be handled by converting them into separate columns using the get_dummies() method. This ensures that each category is represented as a distinct feature, allowing the model to capture the relevant information.

DATASET TUNING

05

- **Splitting The Dataset:** we will remove the target variable (Attrition_Flag) and divide the dataset into two sets. The feature set (X) containing independent variables, and the label set (Y) with the target variable.

- **Normalizing The Independent Variables:** The independent variables are normalized to ensure that all numeric data is scaled uniformly, with a mean of 0 and variance of 1. This normalization process helps prevent any bias that may arise due to different scales or units of measurement.

The Tunning Performed On The Dataset Are As Follows:

- Hyperparameter tuning is performed for two different models: the Random Forest and the Support Vector Classifier (SVC).
- The data is split into training and testing sets using the GridSearchCV method from the sklearn library.
- The hyperparameter in this case is the number of trees to be grown in the random forest (n_estimators), as the optimal number of decision trees is unknown.
- The Pipeline method is used to define the sequence of steps for the training data.
- To address the imbalanced data, we employ SMOTE (synthetic minority oversampling technique) to balance the churn and non-churn data.
- The scoring parameter considered here is precision, aiming to minimize false positives.
- Additionally, a 5-fold cross-validation is applied to the training data.

DATASET TUNING

06

- To determine the best precision score, the model is initially run with *n_estimators* set to [10, 20, 50, 80, 100].
- The precision score achieved is 0.9333 with 100 decision trees.
- Consequently, the model is re-evaluated with *n_estimators* set to [100, 150, 200, 250, 300].
- This time, a slightly improved precision score of 0.9394 is achieved with 150 decision trees.
- Therefore, the optimal number of decision trees in the random forest is determined to be 150.
- In the case of the Support Vector Classifier (SVC), the hyperparameters are the kernel and "c" values.
- After running the SVC, a precision score of 0.97 is obtained with a sigmoid kernel and c = 0.001.
- This indicates that the model requires a hyperplane in 3-dimensional space to separate the churn data.
- However, due to the lower interpretability of SVC, it is difficult to analyze the specific features that strongly contribute to customer churn.

CHOICE OF EVALUATION METRIC

06

True Positive (TP): A true positive occurs when the model correctly predicts an existing customer as an existing customer

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False Positive (FP): A false positive occurs when the model incorrectly predicts a churned customer as an existing customer.

False Negative (FN): A false negative occurs when the model incorrectly predicts an existing customer as a churned customer

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True Negative (TN): A true negative occurs when the model correctly predicts a churned customer as a churned customer.

Given the objective of reducing the number of false positives, precision is the appropriate evaluation metric to focus on. By optimizing for precision, the goal is to minimize the instances where the model incorrectly predicts a customer as an existing customer when they have actually churned. This helps ensure that the model's predictions of existing customers are more accurate and reliable.

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OVERFITTING AVOIDANCE MECHANISM

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- To address overfitting in the Random Forest Classifier (RFC), one approach is to select a subset of features that are deemed important for predicting churn.

- The feature_importances_ method can be used to identify the relative importance of each feature in the RFC model.

- In this project, we experimented with two RFC models: one using the top 5 important features and another using the top 3 important features

- Our objective was to assess if restricting the model to these selected features would improve the precision score.

- Surprisingly, we observed a decrease in precision when reducing the number of features. The precision scores were 0.939 and 0.930 for the models with 5 and 3 features, respectively.

- This outcome suggests that all features in the dataset are important for accurate churn prediction.

- By removing variables, we unintentionally underfit the model since we are providing less data and potentially discarding crucial information.

- Therefore, it is crucial to include all features in the RFC model to maintain its predictive power and avoid compromising its performance.

COMPARISON & RECOMMEND OF MODEL FOR DEPLOYMENT

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After running the random forest classifier, we obtained a precision score of 0.9394 with 180 decision trees using 5-fold cross-validation. On the other hand, the support vector classifier achieved a precision score of 0.9619 with a sigmoid kernel and C value of 0.01 using the same cross-validation technique.

Based on the higher precision score achieved by the support vector classifier (SVC), I would recommend deploying the SVC model for real-world scenarios. The higher precision score indicates that the SVC model performs better at minimizing false positives and accurately predicting existing customers.

It is worth noting that the random forest classifier (RFC) might be underfitting in this case. This could be due to a couple of reasons.

Firstly, the dataset provided may be insufficient for the RFC model to capture the complexity of the underlying function accurately.

Secondly, there might be additional factors not included in the dataset that contribute to customer churn, which the RFC model fails to consider.

These reasons contribute to the underfitting of the RFC model.